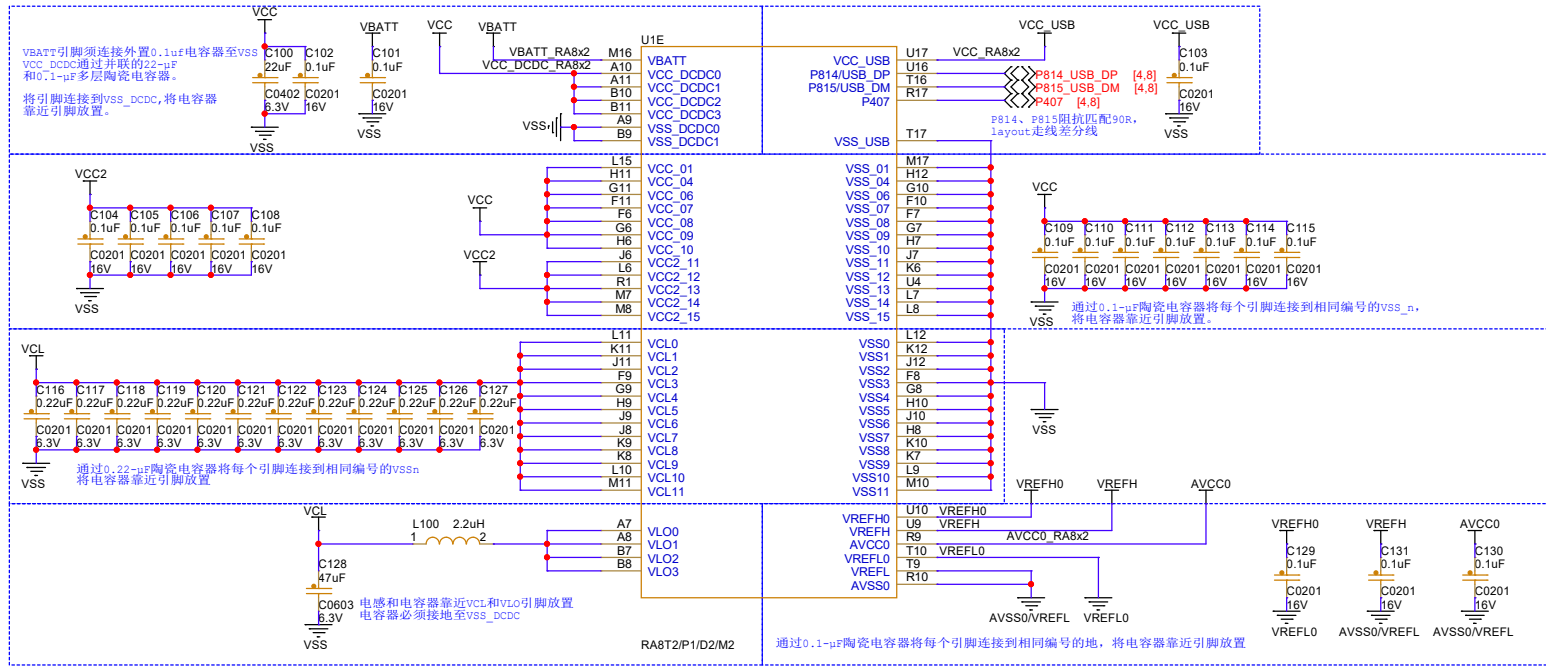




www.gitee.com
/ramcu/cpk_ex
amples



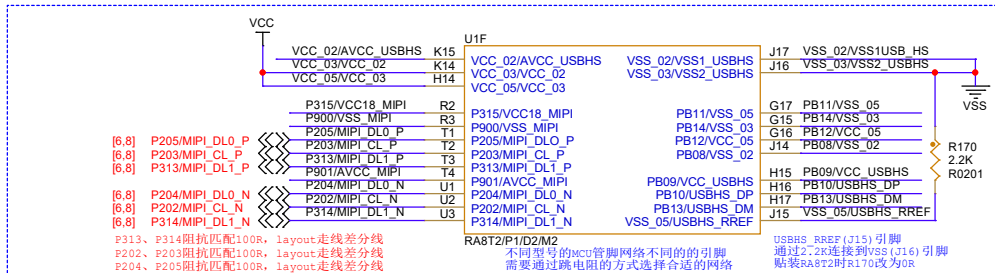
www.github.com
/renesas/cpk_ex
amples



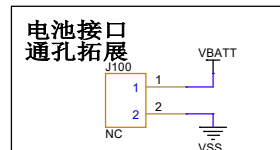
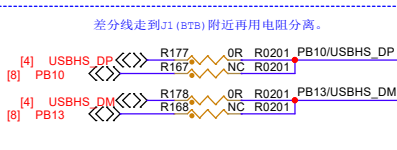
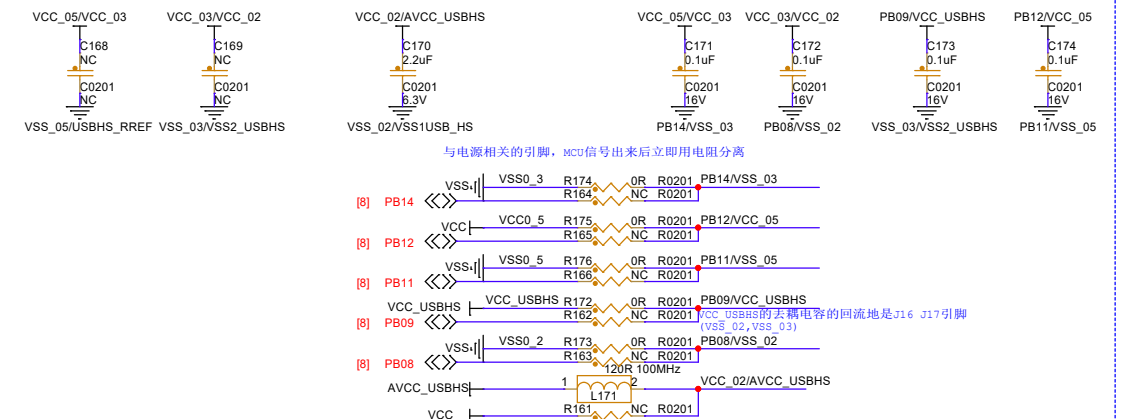
核心板缺省功能阻容设置

核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	0.1uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1uF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R@100M
R172-R180	NC	0R
R191-R193	NC	0R

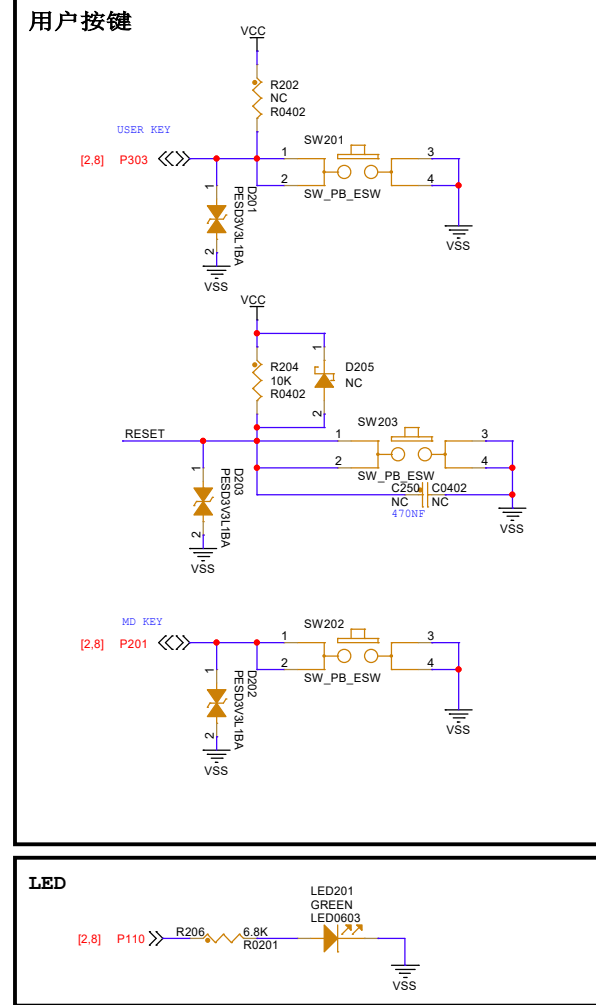
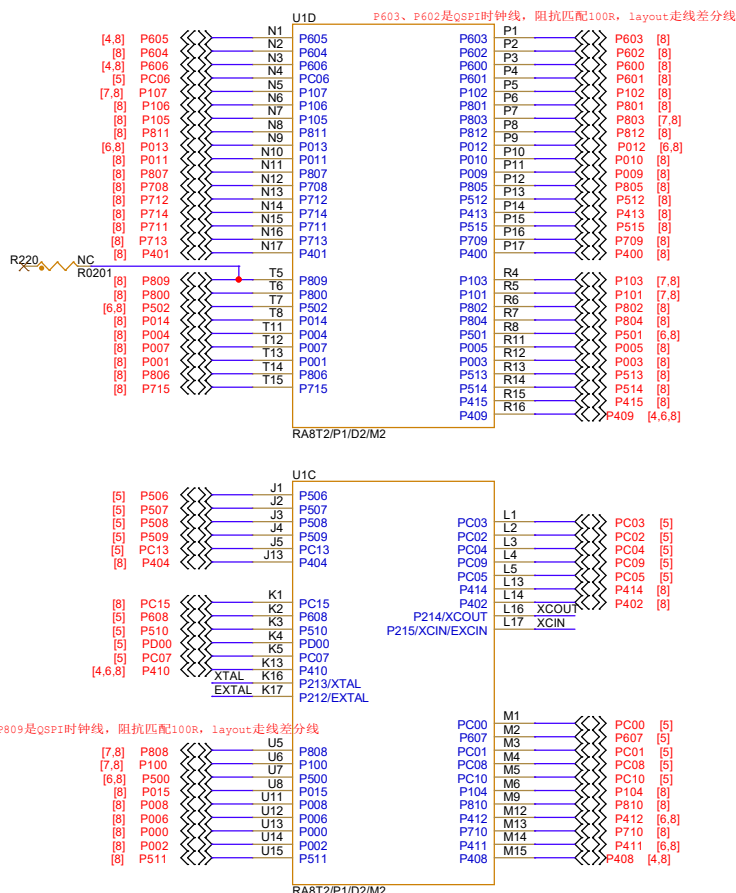
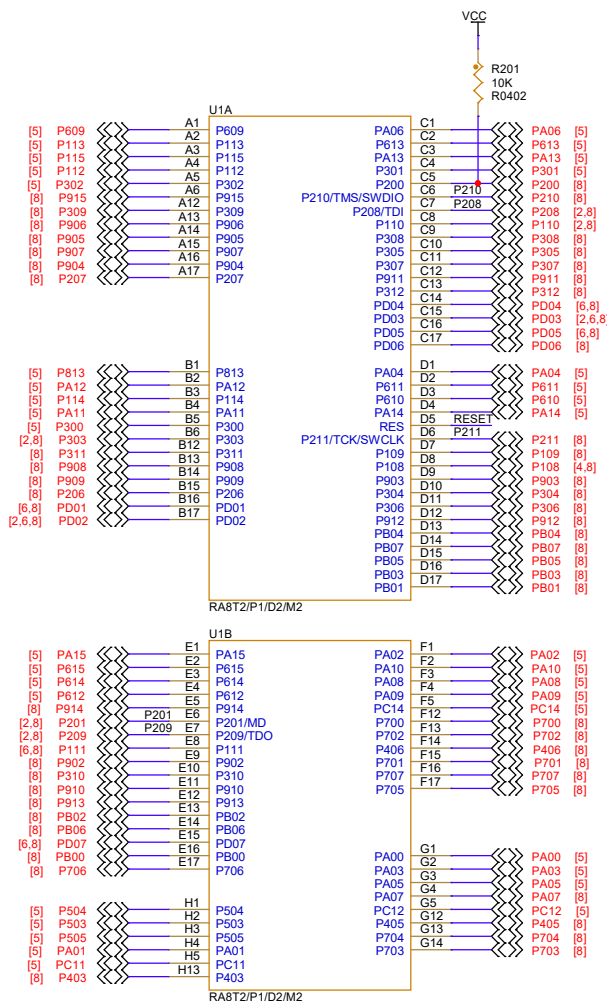
由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。
此原理图默认贴装RA8P1/D2



电阻选装

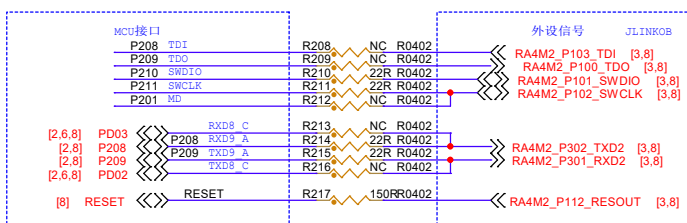


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Title CPKMI-RA8P1核心板			
Size A3	Document Number MCU 电源	Rev V1.1	
Date: Saturday, September 12, 2026	Sheet 1	of 10	



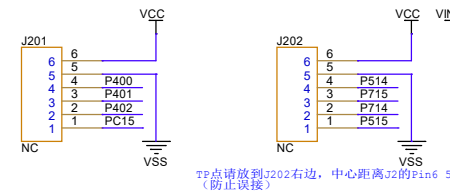
如果使用USB HS，主晶振只能选12/20/24/48MHz

接口信号

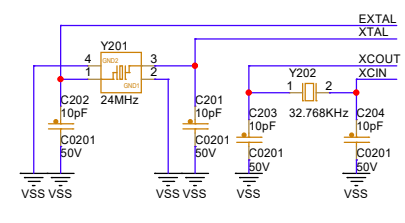


通孔拓展

VIN有大电流需求，请用粗线或铜皮连到VIN网络。
J201, J202之间的间距请保证2.54mm的倍数

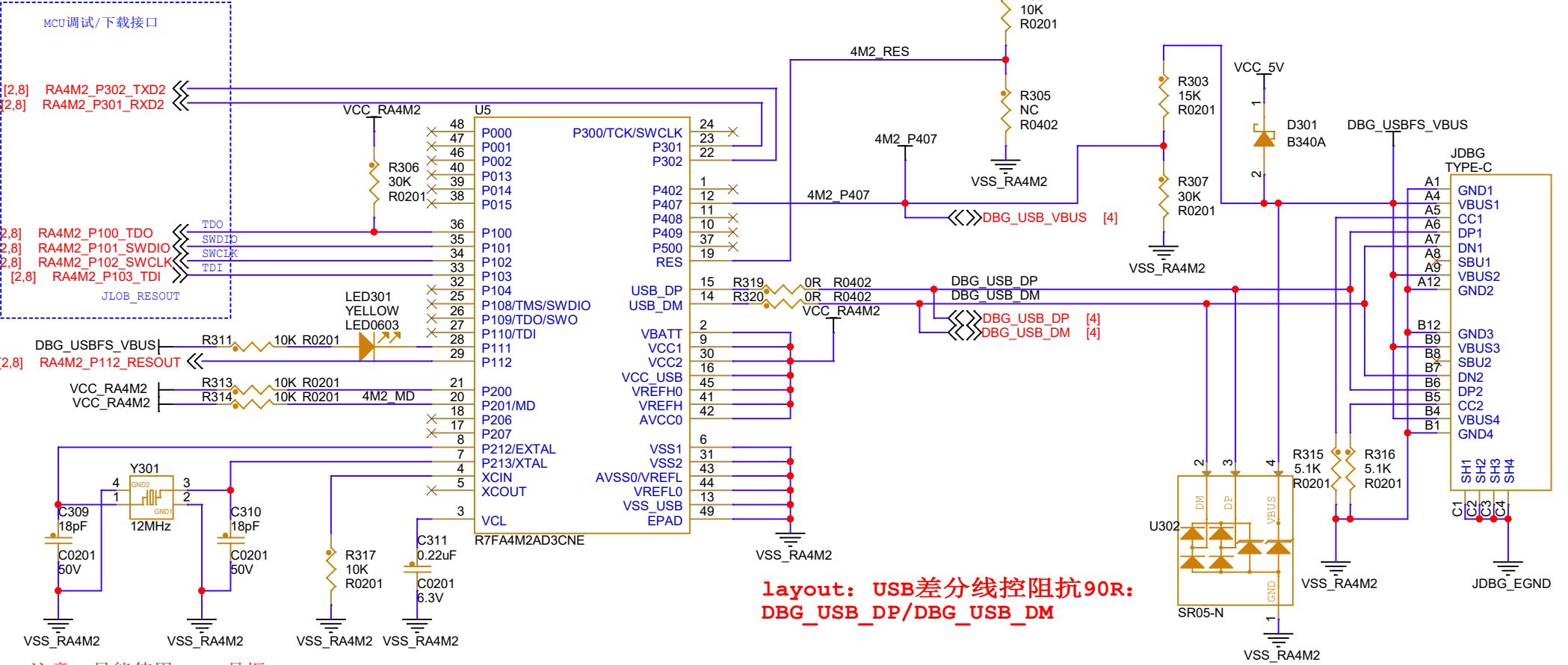


晶振



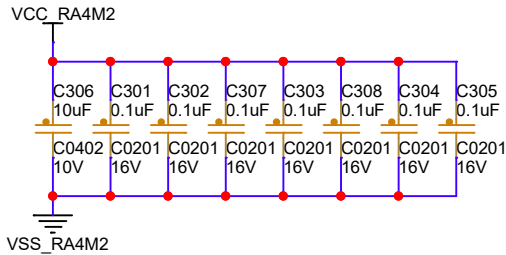
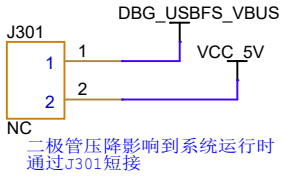
Jlink OB

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- Page4: USB HS
- Page5: SDRAM
- Page6: STORAGE MEMORY
- Page7: 电源LDO/DCDC/MIPI/FLASH
- Page8: BTB



layout: USB差分线控阻抗90R:
DBG_USB_DP/DBG_USB_DM

注意：只能使用12MHz晶振



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如果使用USB HS，主晶振只能选12/20/24/48MHz

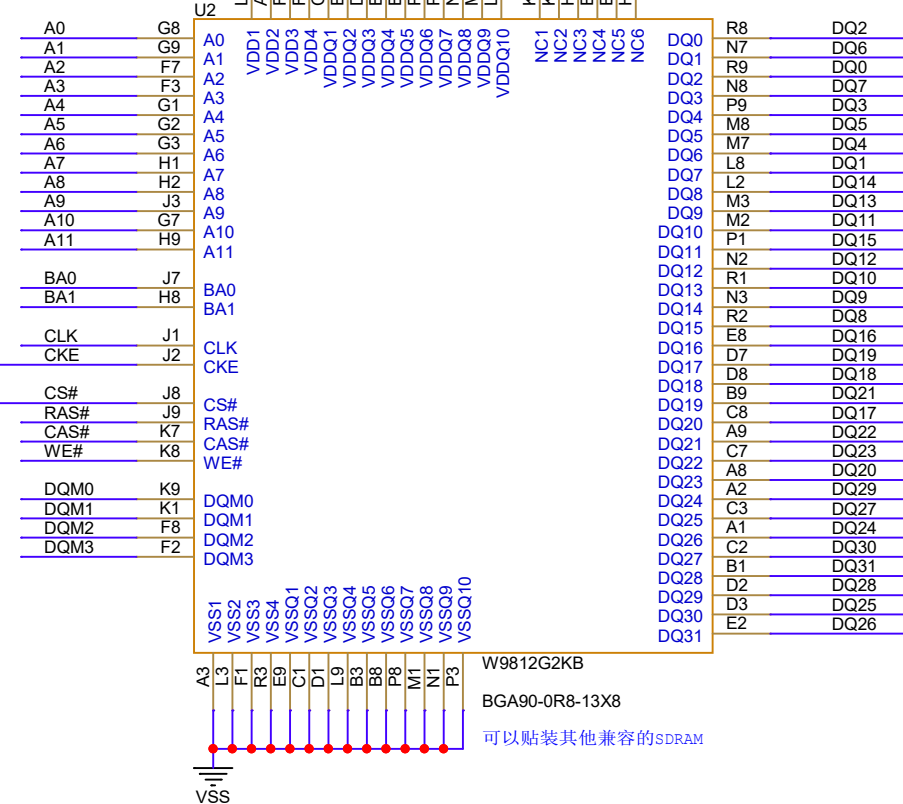
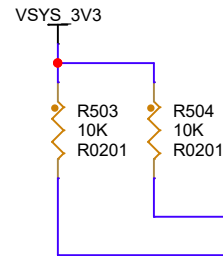
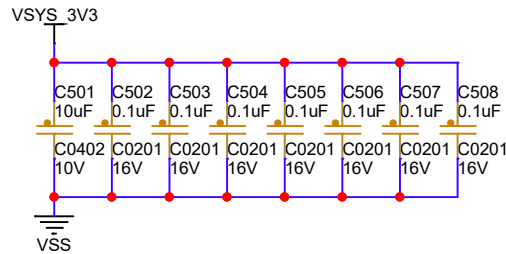
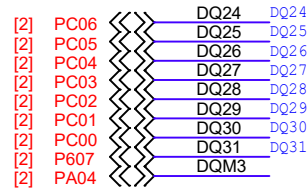
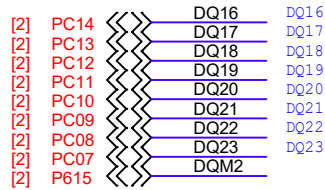
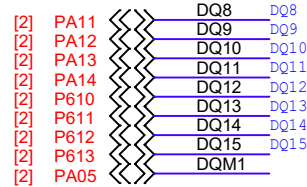
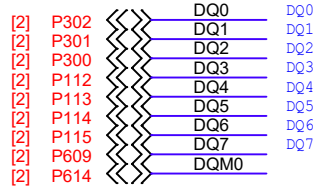
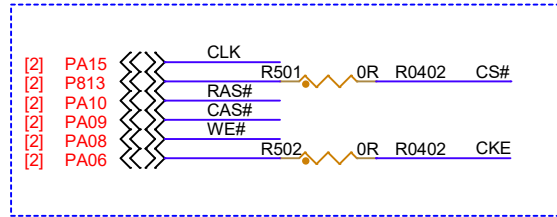
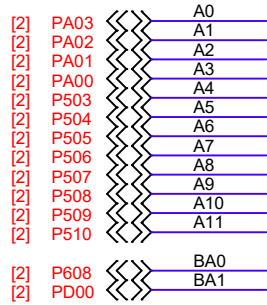


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SDRAM

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layout: SDRAM走线控阻抗50R:



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Title

CPKHMI-RA8P1核心板

Size
A4

Document Number

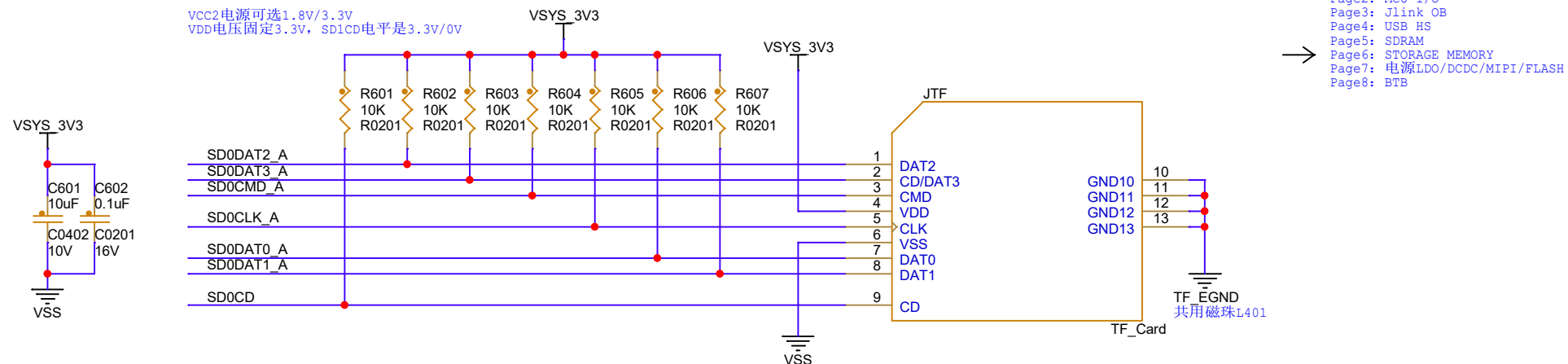
SDRAM

Rev
V1.1

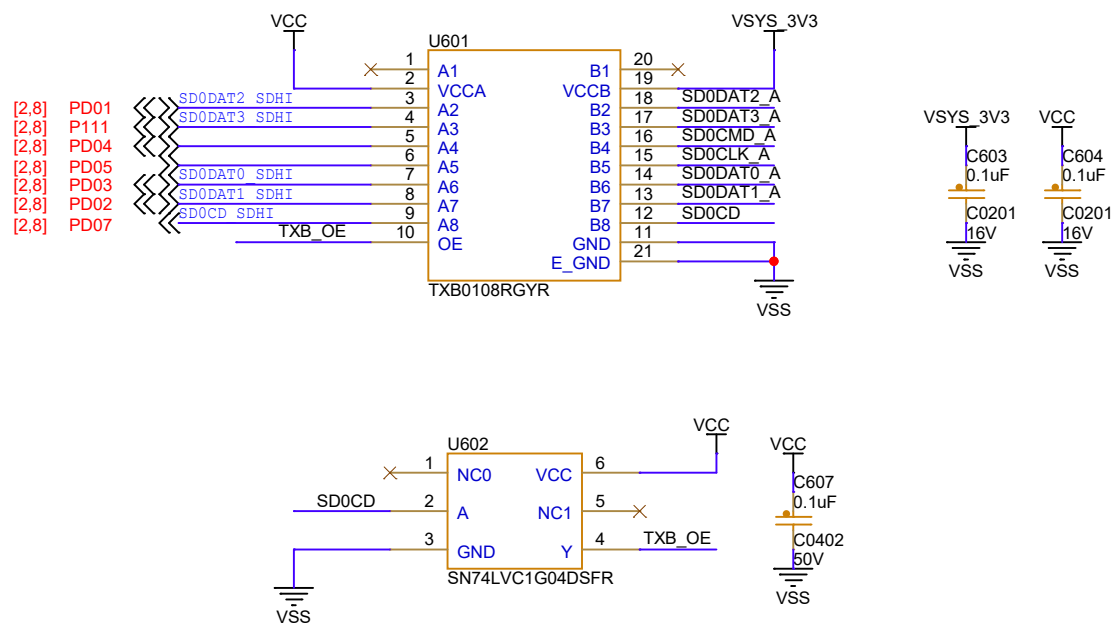
Date: Saturday, September 12, 2026

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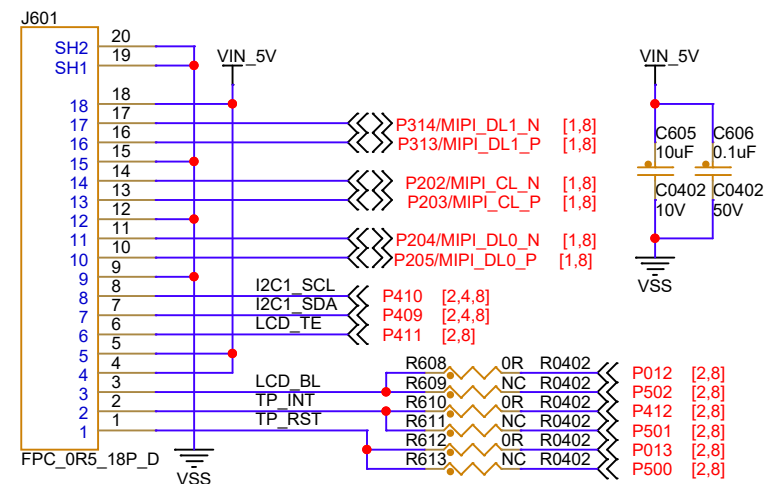
SD卡



SD卡电平转换



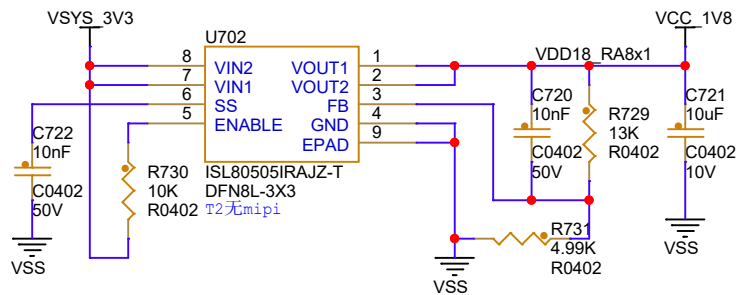
MIPI屏



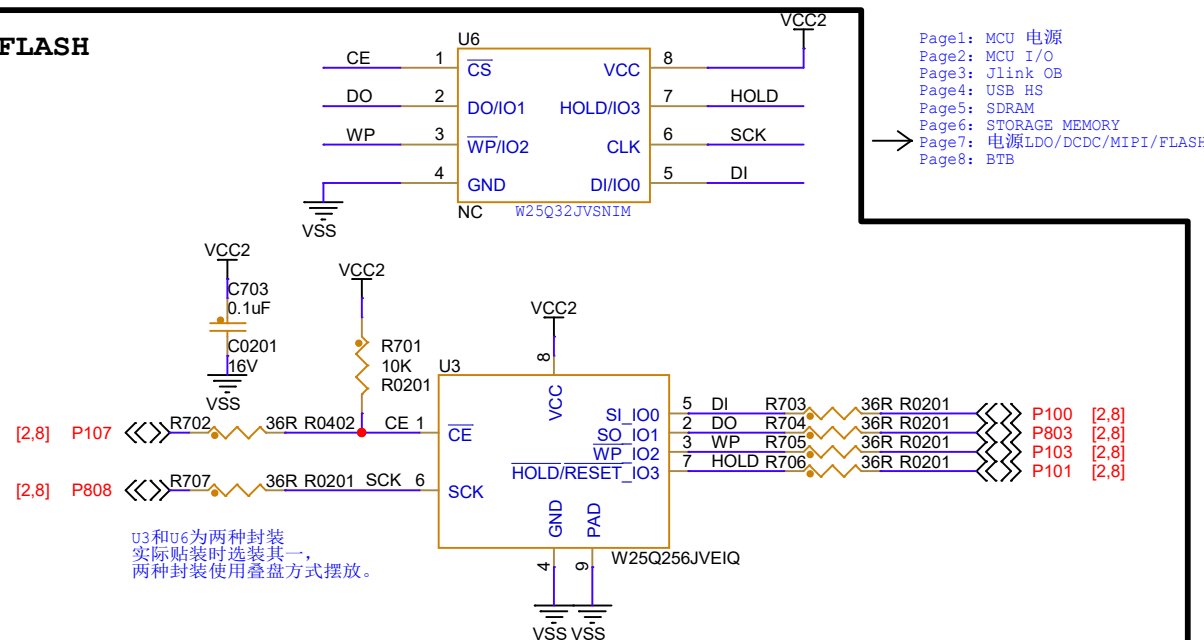
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1.8v电源

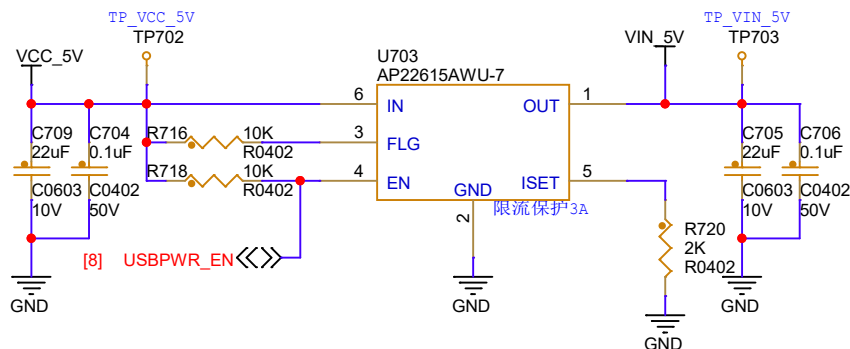


FLASH

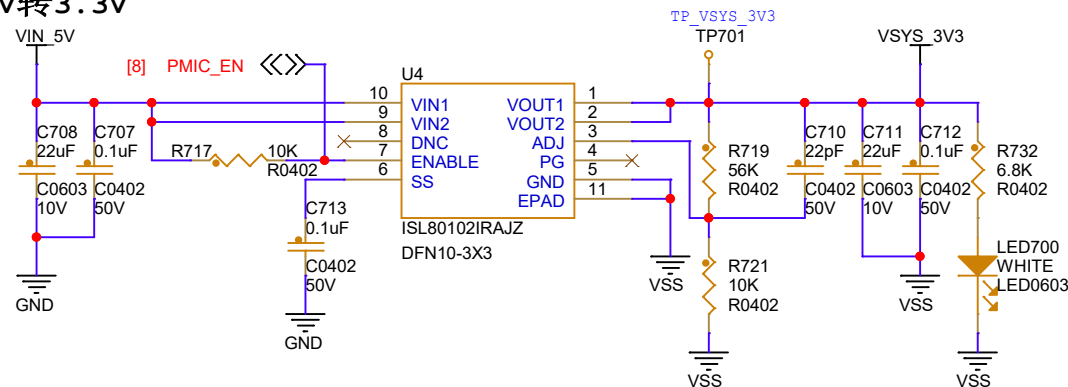


Page1: MCU 电源
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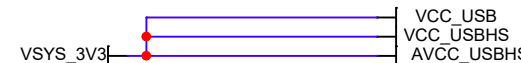
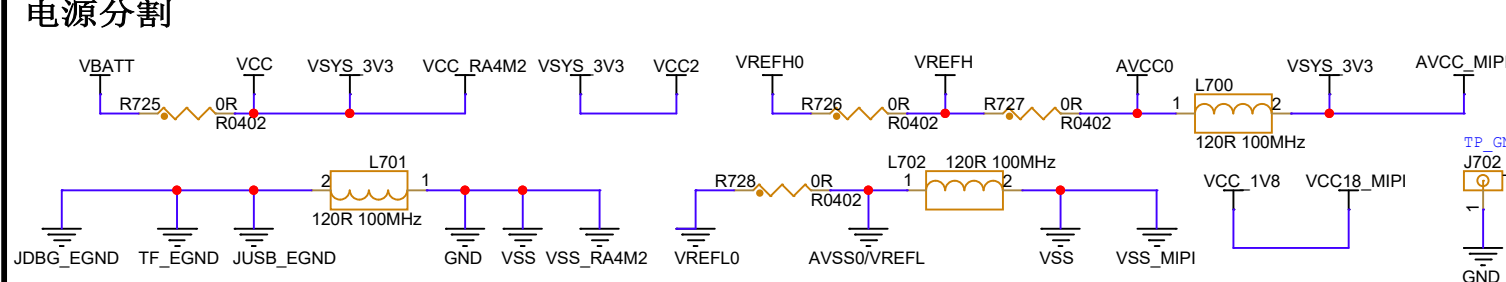
电源保护



5v转3.3v



电源分割

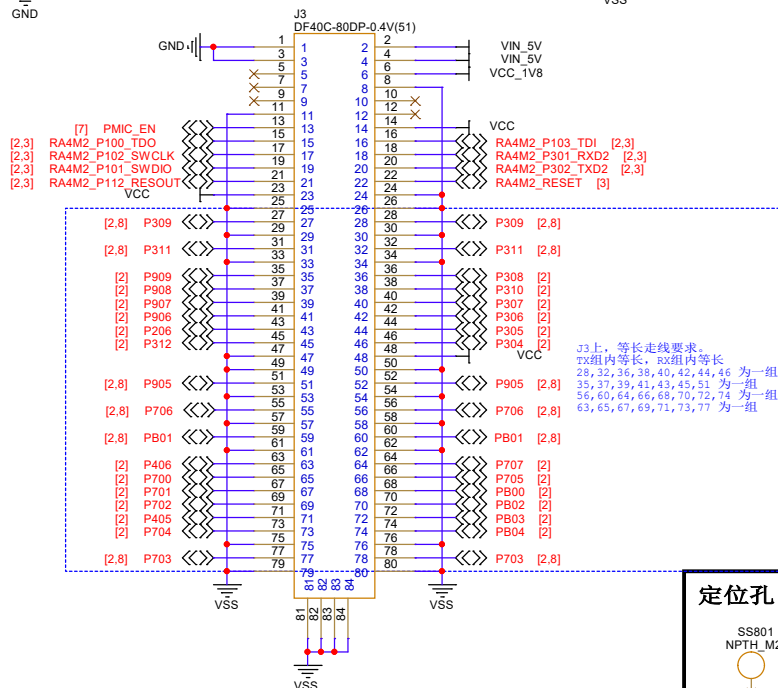
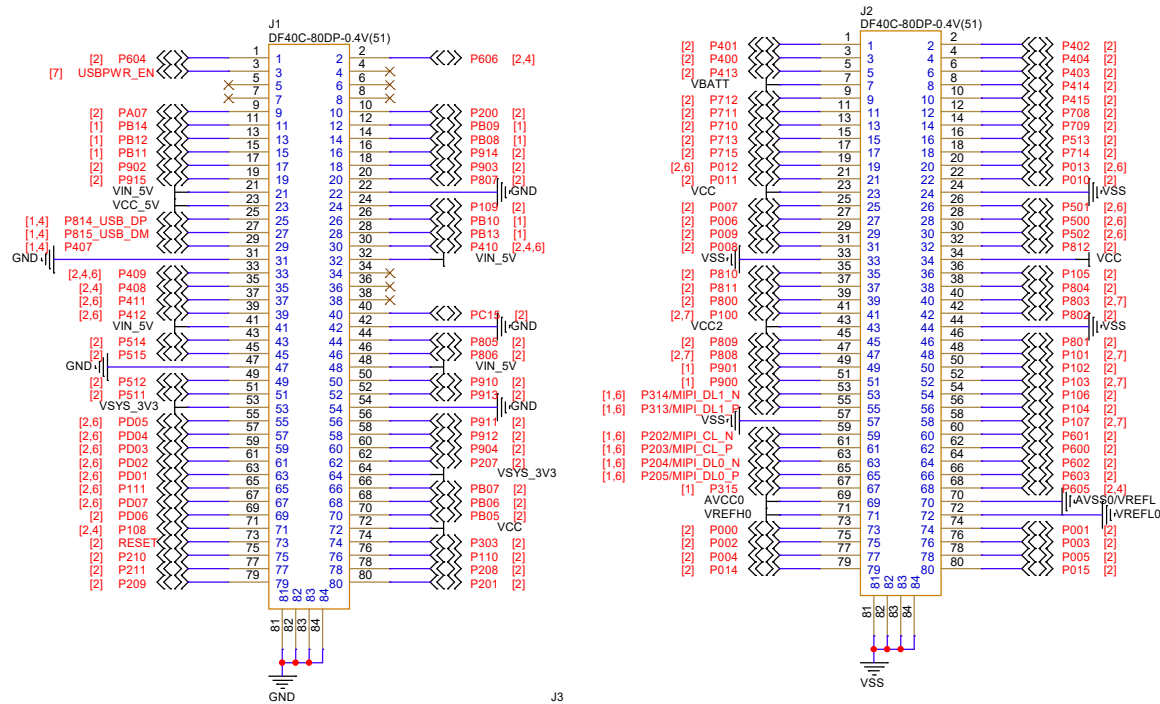


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Title		
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BTB座子

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定位孔



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历史版本

版本号	日期	设计	描述
V1.0	2025-08-28	LPC	量产版本
V1.1	2025-09-17	LPC	修正VCC_USB连接

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